TRUNG V. HA, PhD
Postdoctoral Researcher, Rochester Institute of Technology

Email: vinhtrungha99@gmail.com | ORCID: 0000-0001-6600-2517
GitHub: tvh0021 | LinkedIn: trung-ha-24a16v

Curriculum Vitae

(Last updated: October 15, 2026)

TECHNICAL SKILLS

Programming	Python, C++, MATLAB, MySQL
Packages	NumPy, Pandas, matplotlib, scikit-learn, SciPy, Astropy, JAX
Testing & Deployment	pytest, Sphinx, ReadtheDocs, GitHub Actions
Tools	Linux, Git, Jupyter, VSCode, Slurm/PBS, MPI
ML & Statistics	self-organizing maps, clustering, ensemble methods, hypothesis testing, cross-validation, Monte Carlo resampling, Fourier analysis

RESEARCH PROJECTS

Numerical Simulations of Matter around Supermassive Black Holes

Lead Researcher & Developer 2023 – Present

- Developed numerical solutions in C++ to simulate plasma around supermassive black holes, achieving 100× accuracy improvement against observational data by leveraging novel computational methodologies.
- Ran simulations on distributed supercomputing clusters, leveraging 1000+ CPU cores across MPI-connected networks.
- Secured NSF supercomputing allocation as PI (400,000 CPU hours); managed project timelines and computational resources.
- Analyzed high-dimensional, large-scale (> 10^7 data points) simulation outputs (5000+ snapshots) and reported scientific findings to diverse audiences at 3 international conferences, including an invited seminar at the Vietnam National Space Center.
- Publication:
  - "*Galaxy- and Meso-scale Multiphase Accretion Flows onto a Supermassive Black Hole I: Gas Morphology and Accretion Dynamics*", **Trung Ha** & Yuan Li, 2026 (manuscript in preparation)

Improving Black Hole – Galaxy Correlation with Machine-learning Regressors

ML Advisor & Manuscript Editor 2024 – Present

- Advised a junior researcher on the implementation of ML algorithms (decision tree, random forest, XG Boost, MLP) to improve prediction of black holes mass from their host galaxies properties by over 2× compared to traditional linear methods.
- Supervised the junior researcher on rigorous ML methodologies, including reproducibility pipeline and convergence tests.
- Played a key role in the manuscript writing process, provided feedback and initiated one-on-one editing sessions.
- Publication:

- "*Black Hole-Host Galaxy Correlations with Machine Learning: A Comparative Study of Illustris, TNG, and EAGLE*", Jacob Reinheimer, Yuan Li, **Trung Ha**, 2026, The Open Journal of Astrophysics (in review, https://arxiv.org/abs/2510.20751)

Anomaly Detection in Plasma Turbulence with Machine Learning

Lead Developer & Analyst 2023 – 2025

- Developed and shipped an open-source Python library, *aweSOM*, that performs unsupervised clustering using self-organizing maps and statistical ensembles, trained on dense, multidimensional datasets ($10^7 - 10^8$ rows).
- Engineered convolutional features enabling novel characterization of two previously unresolved structure classes.
- Refactored and optimized legacy codebases, improving memory utilization by 1000×; implemented JAX for GPU acceleration, achieving up to 100× speedup over CPU baseline.
- Designed scaling/unit tests, documented and deployed on ReadtheDocs; adopted by external groups as a production tool.
- Delivered 4 presentations at international conferences in the US and Europe and an invited departmental seminar.
- Publications:
  - "*Machine Learning Characterization of Intermittency in Relativistic Pair Plasma Turbulence: Single and Double Sheet Structures*", **Trung Ha** et al. 2025, Astrophysical Journal Letters (impact factor: 11.7), 985, 31 (https://iopscience.iop.org/article/10.3847/2041-8213/add47b), 5 citations
  - "*aweSOM: a CPU/GPU-accelerated Self-organizing Map and Statistically Combined Ensemble Framework for Machine-learning Clustering Analysis*", **Trung Ha** et al. 2025, Journal of Open Source Software, 10(108), 7613 (https://joss.theoj.org/papers/10.21105/joss.07613)

Novel Characterization of Weak Line Quasars' Properties

Lead Analyst & Scientific Coordinator 2020 – 2023

- Led an international team of 22 scientists from five countries (USA, Canada, China, Chile, Israel), synthesized ideas among experts into a unified analysis framework.
- Developed Python analysis pipelines to extract 20+ properties from spectral observations of hundreds of quasars.
- Migrated legacy MATLAB codebase to Python; reproducible workflow subsequently adopted as group standard.
- Reported novel findings of a non-linear correlation between quasars' accretion rate and their UV spectral signal, disproving a widely assumed break in quasars classification.
- Publications:
  - "*Shedding New Light on Weak Emission-line Quasars in the C IV–Hβ Parameter Space*", **Trung Ha** et al. 2023, Astrophysical Journal (IF: 5.4), 950, 97 (https://iopscience.iop.org/article/10.3847/1538-4357/acd04d), 10 citations
  - "*Gemini Near Infrared Spectrograph−Distant Quasar Survey: Rest-frame Ultraviolet−Optical Spectral Properties of Broad Absorption Line Quasars*", Harum Ahmed, …, **Trung Ha** et al.

2024, Astrophysical Journal (IF: 5.4), 968, 77 (https://iopscience.iop.org/article/10.3847/1538-4357/ad3e69), 7 citations

Large-scale Statistical Inference on Multi-dimensional Observational Data

Lead Researcher 2020 – 2022

- Built Pandas ETL pipelines to clean, transform, and validate data integrity across millions of observations.
- Implemented resampling methods (Monte Carlo, jackknife, bootstrap) to estimate systematic uncertainties in noisy data.
- Developed a parallel two-point correlation estimator achieving 20× higher computational throughput than existing codebases.
- Delivered 3 presentations in the US and Asia and was interviewed by the American Astronomical Society on publication.
- Publications:
  - "*Turbulence in Milky Way Star-forming Regions Traced by Young Stars and Gas*", **Trung Ha** et al. 2022, Astrophysical Journal (IF: 5.4), 934, 7 (https://iopscience.iop.org/article/10.3847/1538-4357/ac76bf), 41 citations
  - "*Measuring Turbulence with Young Stars in the Orion Complex*", **Trung Ha** et al. 2021, Astrophysical Journal Letters (IF: 11.7), 907, 40 (https://iopscience.iop.org/article/10.3847/2041-8213/abd8c9), 23 citations

Scientific Mentor 2024 – 2025

- Mentored a post-baccalaureate student from zero domain knowledge to first-author publication on multi-scale correlation functions and MCMC simulations of stellar data, expanding coverage of proposed methodology from early works.
- Publication:
  - "*Probing Turbulence, Gravity, Supernovae, and Magnetic Field Effects with the 6D Kinematics of Young Stars in Milky Way Star-forming Regions*", Benjamin Velguth, Yuan Li, **Trung Ha** et al. 2025, Astrophysical Journal (IF: 5.4), 990, 165 (https://iopscience.iop.org/article/10.3847/1538-4357/adf5c0)

WORK & RESEARCH EXPERIENCE

Oct 2026 –	**Postdoctoral Researcher,** Center for Computational Relativity and Gravitation, Rochester Institute of Technology
Sep 2024 – Sep 2026	**Graduate Research Assistant,** University of Massachusetts Amherst
Jun 2021 – Aug 2024	**Graduate Research Assistant,** University of North Texas
Sep 2023 – May 2024	**Research Analyst (Pre-doctoral Fellow) and Guest Researcher,** Center for Computational Astrophysics, Simons Foundation
Aug 2020 – May 2021	**Graduate Teaching Assistant,** University of North Texas
Sep 2018 – May 2020	**Undergraduate Research Student,** Center for Computational Relativity and Gravitation, Rochester Institute of Technology
Sep 2018 – Dec 2019	**Undergraduate Teaching Assistant,** University of Rochester
Jun 2018 – Aug 2018	**Summer Research Intern,** Laboratory for Laser Energetics, University of Rochester

EDUCATION

2024 – 2026	**University of Massachusetts Amherst, Amherst, Massachusetts** PhD in Astronomy – conferred September 2026 Thesis title: "*The Next Generation of Diagnostic Tools for Astrophysical Turbulence*", advisor: Yuan Li GPA: 4.00 / 4.00
2020 – 2024	**University of North Texas, Denton, Texas** PhD candidate – transferred to UMass before receiving degree Master of Science in Physics – conferred May 2022 GPA: 4.00 / 4.00
2017 – 2020	**University of Rochester, Rochester, New York** Bachelor of Science in Physics
2015 – 2017	**Central Arizona College, Coolidge, Arizona** Associate of Science

SELECTED TALKS

Dec 2025	**Stars & Plasma Astrophysics Group Meeting,** Flatiron Institute, New York, NY, USA Title: "Modeling Multiphase Accretion Flows onto M87* with Galaxy- and Meso-scale Simulations"
Jun 2025	**INVITED - Vietnam National Space Center,** Astrophysics Department Seminar, Hanoi, Vietnam Title: "Feedback-driven Multiphase Accretion in M87: Results from Meso-scale Simulations"
Jun 2025	**IAUS 397 – UniversAI: Exploring the Universe with Artificial Intelligence**, Athens, Greece Title: "aweSOM: an Open-source Python Package for Efficient Clustering of Intermittency in Magnetized Plasma Turbulence"
Mar 2025	**Stars & Plasma Astrophysics Group Meeting,** Flatiron Institute, New York, NY, USA Title: "Segmentation of Single and Double Current Sheets in Magnetized Plasma Turbulence with Machine Learning"
Jun 2024	**AstroAI Workshop,** Harvard-Smithsonian Center for Astrophysics, Cambridge, MA, USA, and **Midwest Magnetic Fields Workshop,** Madison, WI, USA Title: "Segmentation of Current Sheets in Magnetized Plasma Turbulence with Computer Vision"
Apr 2024	**INVITED - Center for Computational Relativity and Gravitation Lunch Talk,** Rochester, NY, USA Title: "Can Neural Networks Recognize Current Sheets? Using Computer Vision to Analyze Magnetized Plasma Turbulence"
Feb 2024	**Kavli Institute for Theoretical Physics (KITP) – Turbulence in the Universe Workshop,** Santa Barbara, CA, USA Title: "Segmentation of Current Sheets in Magnetized Plasma Turbulence with Computer Vision"
Dec 2023	**Black Holes on Broadway: The Next Generation of AGN Models in Galaxy Formation,** New York, NY, USA

	Title: "Bridging the Gap: Modeling Supermassive Black Holes Feeding and Feedback at the Meso-scale"
Aug 2022	**Star Formation in Different Environments 2022,** Rencontres du Vietnam, Quy Nhon, Vietnam Title: "Turbulence in Milky Way Star-forming Regions Traced by Young Stars and Gas"

PROFESSIONAL SERVICE

Served as referee for the Astrophysical Journal Letters.
Mentored two undergraduate students (Alexis NieMeira, Jonathan Red), one post-baccalaureate fellow (Benjamin Velguth), and two junior graduate students (Jacob Reinheimer, Roy Hsu), resulting in two publications (Velguth et al. 2025 and Reinheimer et al. 2026, as third author).

SUPERCOMPUTING & OBSERVING AWARDS

Dec 2024	**Co-I., XMM-Newton proposal** Title: "Identifying a Robust and Practical Accretion-Rate Indicator for Distant Quasars"; observing time: 110 kiloseconds.
Mar 2024	**P.I., NSF ACCESS Explore allocation** Title: "Turbulent multiphase accretion flows from supermassive black hole feedback"; amount: 400,000 CPU hours

AWARDS AND HONORS

Spring 2025	Mary Dailey Irvine Graduate Travel Grant, UMass Amherst, $1000
2023 – 2024	The Zhibing Hu Scholarship, University of North Texas, $1000.
May 2023	Featured on University of North Texas College of Science news: "UNT Physics Graduate Student Selected for Prestigious Pre-Doctoral Program" (link to article)
Spring 2023	College of Science Travel Award, University of North Texas. $500.
Feb 2022	Featured on University of North Texas's Graduate Research Spotlight.
2021 – 2025	R. B. Toulouse Scholarship, University of North Texas. $1000/year.
Apr 2021	Featured on the *North Texan*: "Turbulent Motion Moves Research Forward" (link to article).
2019 – 2020	Take Five Scholar, University of Rochester. Thesis: "Exploring the Advantages and Shortcomings of French Literature in Translation".

OTHER ACTIVITIES

Participated in the Flatiron Institute's Center for Computational Astrophysics Pre-doctoral program in New York City in fall 2023.
Organizer for the weekly joint-UNT/UTD astronomy journal club, 2023.
Interviewed on AAS Journal Author Series about publication titled: "Measuring Turbulence with Young Stars in the Orion Complex" on YouTube, 2021.
Participated in student exchange programs: "Cultural Exchange Program" in Arizona, USA during the 2014-15 school year and "French in France" in Rennes, France in summer 2019.

Citizenship: Vietnam
U.S. Permanent Resident

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GitHub: tvh0021 | LinkedIn: trung-ha-24a16v

Publication List

In reverse chronological order, refereed publications are listed first

FIRST AUTHOR PUBLICATIONS

1. ***"Machine-Learning Characterization of Intermittency in Relativistic Pair Plasma Turbulence: Single and Double Sheet Structures"***
   Ha, Trung; Nättilä, J.; Davelaar, J.; Sironi, L. (2025ApJ...985L..31H)
2. ***"aweSOM: a CPU/GPU-accelerated Self-organizing Map and Statistically Combined Ensemble Framework for Machine-learning Clustering Analysis"***
   Ha, Trung; Nättilä, J.; Davelaar, J. (2025.JOSS.07613, GitHub)
3. ***"Shedding New Light on Weak Emission-Line Quasars in the CIV–Hβ Parameter Space"***
   Ha, Trung; Dix, C.; Matthews, B. M.; Shemmer, O.; et al., (2023ApJ...950...97H)
4. ***"Turbulence in Milky Way Star-forming Regions Traced by Young Stars and Gas"***
   Ha, Trung; Li, Y.; Kounkel, M.; Xu, S.; Li, H.; Zheng, Y., (2022ApJ...934....7H)
5. **"*Measuring Turbulence with Young Stars in the Orion Complex*"**
   Ha, Trung; Li, Y.; Xu, S.; Kounkel, M.; Li, H., (2021ApJ...907L..40H)
6. ***"Galaxy- and Meso-scale Multiphase Accretion Flows onto a Supermassive Black Hole I: Gas Morphology and Accretion Dynamics"***
   Ha, Trung; Li, Y., (in prep.)

OTHER PUBLICATIONS

7. ***"Probing Turbulence, Gravity, Supernovae, and Magnetic Field Effects with the 6D Kinematics of Young Stars in Milky Way Star-forming Regions"***
   Velguth, B.; Li, Y.; **Ha, Trung**, et al. (2025ApJ...990..165V)
8. ***"Rest-Frame Optical Spectroscopy of Ten z ~ 2 Weak Emission-Line Quasars"***
   Chen, Y.; …; **Ha, Trung**, et al. (2024ApJ...972..191C)
9. ***"Gemini Near Infrared Spectrograph - Distant Quasar Survey: Rest-Frame Ultraviolet-Optical Spectral Properties of Broad Absorption Line Quasars"***
   Ahmed, H.; …; **Ha, Trung**, et al., (2024ApJ...968...77A)
10. ***"The Nature of the Motions of Multiphase Filaments in the Centers of Galaxy Clusters"***
    Ganguly, S.; …; **Ha, Trung**, (2023FrASS..1038613)
11. ***"Handing-Off the Outcome of Binary Neutron Star Mergers for Accurate and Long-Term Post-Merger Simulations"***
    Lopez Armengol, F. G.; …; **Ha, Trung**; et al., (2022PhRvD.106h3015L)
12. ***"HARM3D+NUC: A new method for simulating the post-merger phase of binary neutron star mergers with GRMHD, tabulated EOS and neutrino leakage"***
    Murguia-Berthier, A.; …; **Ha, Trung**, et al., (2021ApJ...919...95M)
13. ***"Black Hole-Host Galaxy Correlations with Machine Learning: A Comparative Study of Illustris, TNG, and EAGLE"***
    Reinheimer, J.; Li, Y.; **Ha, Trung**, et al. (submitted to OJA, arXiv:2510.20751)